

Department of Information Technology

Title: “SkillBridge – A Smart Platform for Hiring Verified Skilled Professionals”

Project Review I

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Problem Statement

Millions of skilled workers in India struggle to find consistent work due to lack of digital access, poor visibility, and dependence on middlemen. At the same time, people and businesses find it difficult to hire verified, nearby service providers. Most platforms today focus only on urban or white-collar jobs, leaving skilled trade workers behind. There's a clear gap between skill availability and service demand. SkillBridge aims to bridge this gap with a smart, AI-powered platform that connects skilled individuals directly with real job opportunities.

Objectives

The primary goal of the SkillBridge project is to create a functional and accessible digital ecosystem that empowers skilled manual workers and streamlines their connection with job providers. The project focuses not only on job discovery but also on promoting trust, transparency, and inclusivity in the informal labor sector

To achieve this, the specific objectives of the project are outlined below:

Objective 1 – Requirement Analysis & Platform Foundation To design and develop a digital platform that connects skilled manual workers with local job providers, reducing communication gaps and worker idle time.

Objective 2 – Core Functionality & User Experience

To implement essential system features such as skill-based categorization, location-based search, job-type filtering, and real-time availability to enhance usability and efficiency.

Objective 3 – Trust & Reliability Mechanisms

To establish credibility within the system through profile verification, rating mechanisms, and review systems, ensuring reliable and transparent worker-employer interactions.

Objective 4 – Intelligent System Enhancement Using AI/ML

To integrate AI and Machine Learning techniques for personalized job recommendations, feedback sentiment analysis, and dynamic trust scoring that improves over time.

Objective 5 – Accessibility, Scalability & Inclusivity

To ensure the platform is accessible to users with low digital literacy through multilingual support, icon-driven and voice-assisted interfaces, while maintaining scalability for future regional and sectoral expansion.

Literature Review

St.No	Paper Title	Publisher & Year	Drawbacks
1	PLFS Annual Report	National Statistical Office (NSO), Government of India, 2023	Data collection and release often have delays, reducing real-time policy relevance.
2	AI-Driven Labor Platforms	IJCA, 2021	Doesnt address how AI models will handle real-world labor complexities like skill mismatch
3	NITI Aayog Report on India's Gig and Platform Economy	NITI Aayog, Government of India, 2023	Doesn't fully explore state-level or city-level dynamics of the gig workforce.

St.No	Paper Title	Publisher & Year	Drawbacks
4	Digital Platforms for Informal Labor in India: A Case Study	Journal of Socio-Economic Development, 2022	Focuses on one case study; lacks AI automation and rural scalability.
5	Mobile-First Solutions for Rural Employment Access	International Journal of ICT and Development, 2021	Covers UI design but lacks intelligent job matching or platform automation.
6	AI in Blue-Collar Service Platforms: Challenges and Opportunities	AI & Society, 2020	Lacks implementation details; mostly theoretical and high-level analysis.

Proposed Methodology

Methodology Overview The SkillBridge system is developed using a phased and agile-based approach to ensure scalability, usability, and continuous improvement. The platform follows iterative development with regular validation and feedback integration.

Development Approach

- Requirement Analysis & Planning
- UI/UX Design (User-Centric & Accessible)
- Frontend Development

- Backend Development
- Database Integration
- AI/ML Integration
- Testing & Validation

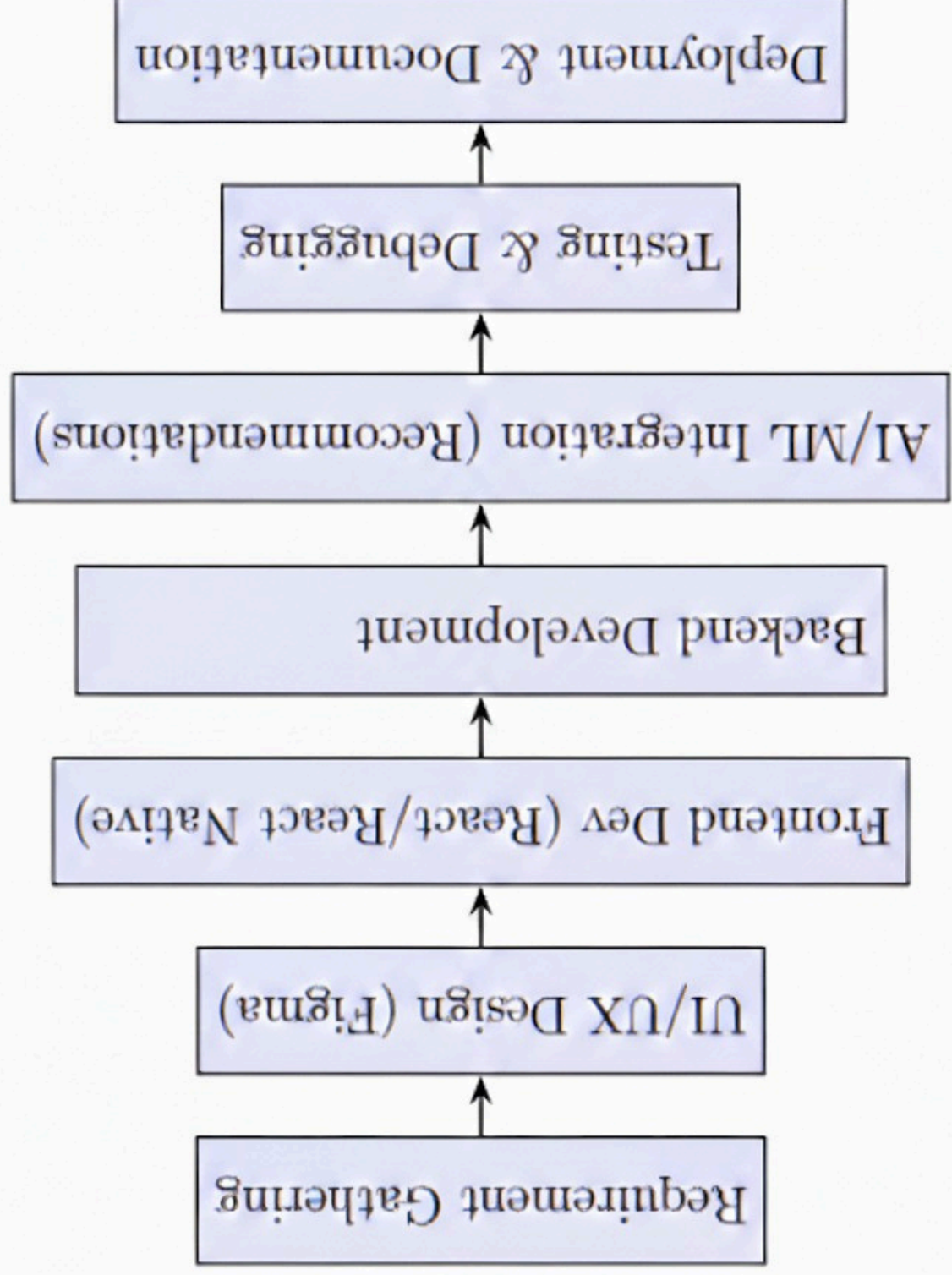
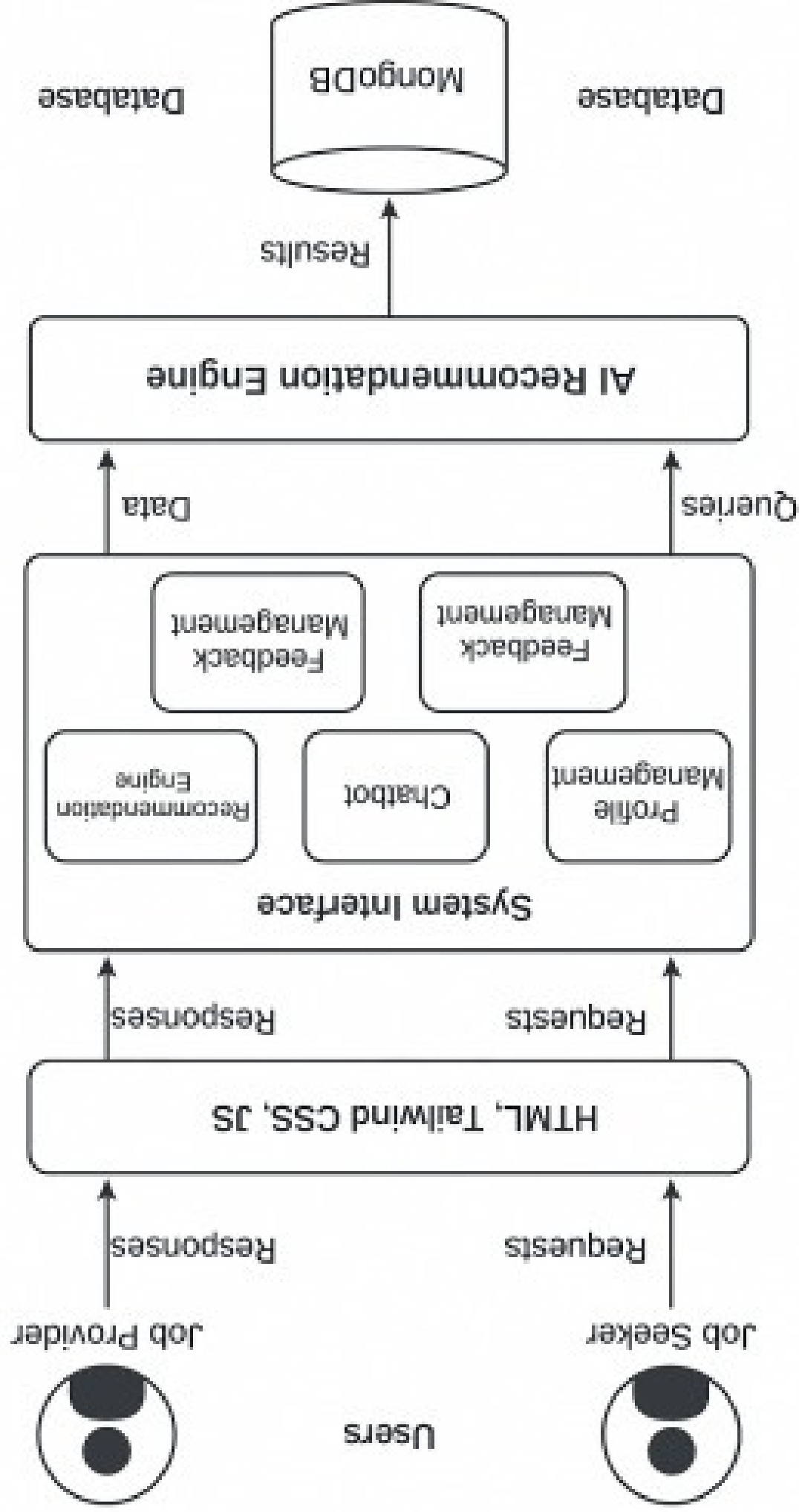


Figure 1: Project Development Flowchart

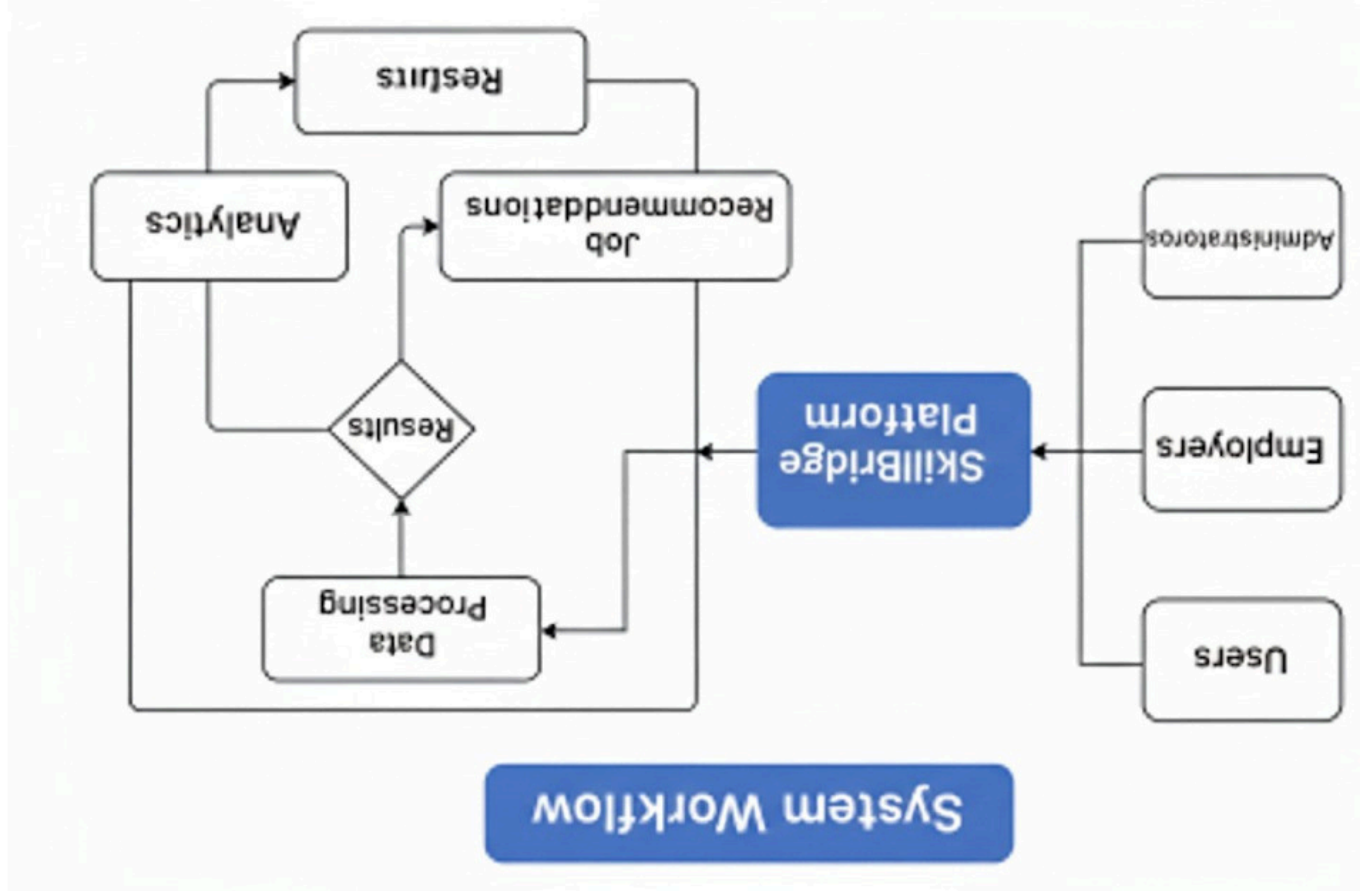
System Architecture :



Phase wise Implementation Planning

Months Activities	Jan 26	Feb 26	March 26	April 26
Problem Statement Identification	✓			
Literature Review	✓			
Requirement Analysis	✓	✓		
Designing		✓		
Experimental Analysis		✓	✓	
Module Wise Implementation		✓	✓	
Testing And Debugging				✓
Project Report Preperation				✓

Use Case / System Flow



Conclusion

Skillbridge demonstrates the effective use of artificial intelligence in a service-based marketplace by connecting skilled workers with customers efficiently. Through AI-powered recommendations, location-based matching, sentiment-based feedback analysis, dynamic pricing, and secure face-recognition authentication, the platform offers a reliable, transparent, and smarter alternative to traditional service-matching systems.

References

- [1] Ministry of Labour and Employment, Government of India, “Periodic Labour Force Survey (PLFS) 2022–23 Annual Report,” New Delhi, India, 2023.
- [2] R. Kaur and P. Singh, “AI-driven job platforms in India,” *Int. J. Comput. Appl. (IJCA)*, vol. XX, no. X, pp. XX–XX, 2021.
- [3] NITI Aayog, “India’s gig economy outlook,” Government of India, New Delhi, India, 2023.

4] A. Mahajan and R. Gupta,
“Digital platforms for informal labor in India: A case study,”
J. Socio-Economic Development, vol. XX, no. X, pp. XX–XX, 2022.

[5] S. Verma and P. Kulkarni,
“Mobile-first solutions for rural employment access,”
Int. J. ICT Develop., vol. XX, no. X, pp. XX–XX, 2021.

[6] T. Sharma and N. Bose,
“AI in blue-collar service platforms: Challenges and opportunities,”
AI & Society, vol. XX, no. X, pp. XX–XX, 2020.

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